Experiment No. 8

Sliding Window Protocol (Go-Back-N)

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Simulation of the Sliding Window Protocol (Go-Back-N variant) using a C program

Objective:

- To understand how the Sliding Window Protocol controls the flow of frames between sender and receiver.
- To simulate the Go-Back-N ARQ variant using arrays and simple logic in C.
- To observe how the sender window slides forward when acknowledgments are received.

Software/Tools Required:

- Linux (Ubuntu) system
- GCC compiler
- Text editor (nano / vim) and Terminal

Step-by-step Procedure:

Algorithm: Sliding Window Protocol (Go-Back-N ARQ)

Input: total number of frames to send N, and window size W.

Sender side

- Initialize base = 0 (first frame in window) and next_seq_num = 0 (next frame to send).
- While base < N:
  - While next_seq_num < base + W and next_seq_num < N: send frame next_seq_num and increment next_seq_num.
  - Wait for acknowledgment. If the ACK for frame base is received, increment base.
  - Otherwise (timeout or error), retransmit all frames from base up to next_seq_num - 1.
- Repeat until all frames are sent and acknowledged (base >= N).

Receiver side

- Initialize expected_seq_num = 0.
- When a frame with sequence number seq_num arrives:
  - If seq_num == expected_seq_num: accept the frame, deliver it to the higher layer, send ACK for expected_seq_num and increment expected_seq_num.
  - Else: discard the frame and resend ACK for expected_seq_num - 1.

Part A: Writing, Compiling and Running the Program

Step 1: Write the program

- Create the file using an editor and save it as `sliding_window279.c`.

Step 2: Compile the program

```
gcc sliding_window279.c -o sliding_window279
```

Step 3: Run the program

```
./sliding_window279
```

Step 4: Provide input and observe

- Enter the number of frames (max 50) and the window size.
- The sender displays the current window and sends all frames inside it.
- The receiver is simulated: each frame is received and an ACK is printed.
- Enter **y** if all frames were received correctly - the window slides forward by the window size.
- Enter **n** to simulate an error - the sender resends from the base frame (Go-Back-N).

Observation Table

Step	Command / Action	Observation
1	gcc sliding_window279.c -o sliding_window279	Program compiled successfully
2	./sliding_window279 with 5 frames, window size 3	Window [0] [1] [2]; frames 0, 1, 2 sent
3	Receiver simulation	ACKs 1, 2, 3 sent for frames 0, 1, 2
4	Input y (all frames received correctly)	Window slid to [3] [4]; frames 3, 4 sent and ACKed
5	Input y again	All frames sent successfully

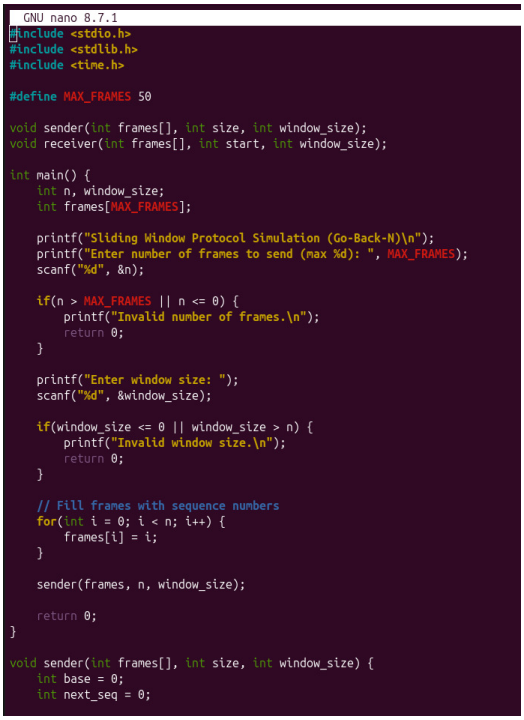
Result:

- Successfully simulated the Sliding Window Protocol (Go-Back-N) in C.
- Observed the sender window sliding forward as acknowledgments were received.

Conclusion:

- Go-Back-N allows the sender to transmit multiple frames before waiting for ACKs.
- On an error, retransmission starts again from the frame where the error occurred.
- It is efficient for networks with fewer errors and longer delays.

Output / Screenshots



```
 GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

#define MAX_FRAMES 50

void sender(int frames[], int size, int window_size);
void receiver(int frames[], int start, int window_size);

int main() {
    int n, window_size;
    int frames[MAX_FRAMES];

    printf("Sliding Window Protocol Simulation (Go-Back-N)\n");
    printf("Enter number of frames to send (max %d): ", MAX_FRAMES);
    scanf("%d", &n);

    if(n > MAX_FRAMES || n <= 0) {
        printf("Invalid number of frames.\n");
        return 0;
    }

    printf("Enter window size: ");
    scanf("%d", &window_size);

    if(window_size <= 0 || window_size > n) {
        printf("Invalid window size.\n");
        return 0;
    }

    // Fill frames with sequence numbers
    for(int i = 0; i < n; i++) {
        frames[i] = i;
    }

    sender(frames, n, window_size);

    return 0;
}

void sender(int frames[], int size, int window_size) {
    int base = 0;
    int next_seq = 0;
```

Fig 1: Code (part 1)



```
    char choice;

    srand(time(NULL));

    while(base < size) {
        printf("\nSender window: ");

        for(int i = base; i < base + window_size && i < size; i++) {
            printf("[%d] ", frames[i]);
        }

        printf("\nSending frames...\n");

        for(int i = base; i < base + window_size && i < size; i++) {
            printf("Sent frame: %d\n", frames[i]);
        }

        printf("\nSimulating receiver...\n");
        receiver(frames, base, window_size);

        printf("\nDid all frames receive correctly? (y/n): ");
        scanf(" %c", &choice);

        if(choice == 'y' || choice == 'Y') {
            base += window_size;
        } else {
            printf("Resending from frame %d...\n", frames[base]);
            // In Go-Back-N, we resend from base
        }
    }

    printf("\nAll frames sent successfully!\n");
}

void receiver(int frames[], int start, int window_size) {
    for(int i = start; i < start + window_size; i++) {
        printf("Received frame: %d, sending ACK: %d\n",
               frames[i], frames[i] + 1);
    }
}
```

Fig 2: Code (part 2)

Output / Screenshots

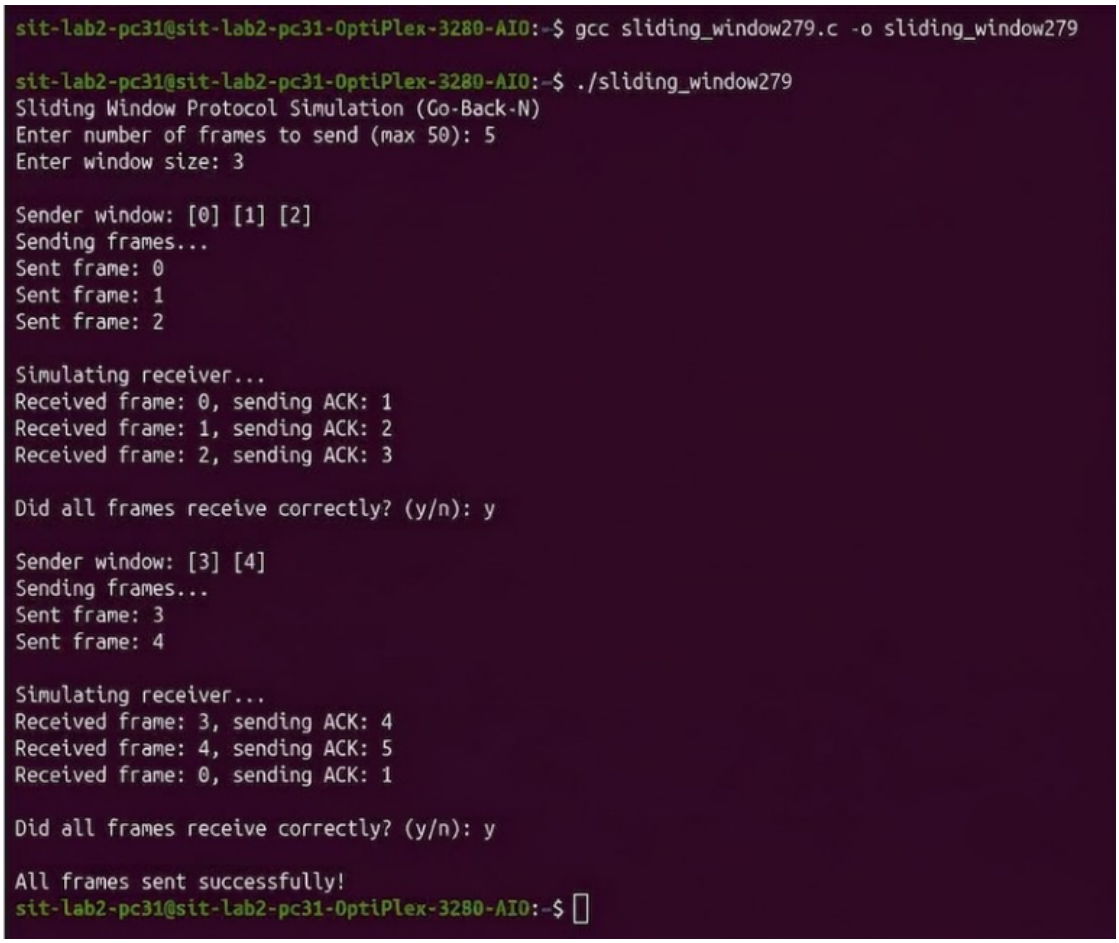


Fig 3: Compilation and output